Exp no: 04

Date: 5/8/26

Setting up a Local Area Network

Aim:

To design, setup and configure a LAN connecting 4 component through a network switch, assign static IP address to all devices, verify connectivity using ping command and demonstrate resource or data sharing across network. demonstrate resource or d

Equipment Required:

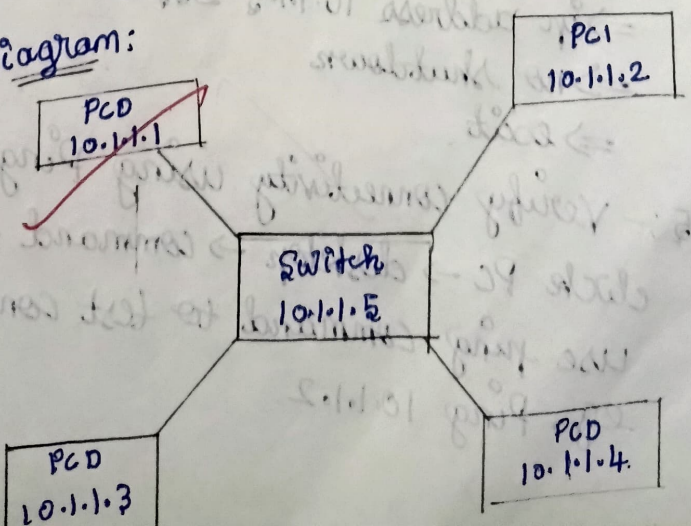
1	Personal computer (PC)	4	Ethernet (NIC) enabled
2	Network Switch	1	8/16/24 port unmanaged/ managed
3	Ethernet Cables (Cat 5e/Cat 6)	4	straight-through RJ-45 connectors.

Theory:

⇒ LAN is a network that connects computer and devices within limited area like home, other...

⇒ Star Topology → all end devices are individually connected to central switch.

Network Diagram:



IP	Address	configuration	Table
Pc0	10.1.1.1	255.0.0.0	Switch Port Fa 01
Pc1	10.1.1.2	255.0.0.0	Switch Port Fa. 0
Pc2	10.1.1.3	255.0.0.0	Switch Port Fa 0
Pc3	10.1.1.4	255.0.0.0	Switch Port Fa
Switch	10.1.1.5	255.0.0.0	VLAN 1 Enbo

Implementation of Cisco Packet Tracket:-

Physical hardware was limited, the LAN was also implemented and verified using cisco packet Tracer simulation Software following the steps below:-

Step 1: Place the devices.

open cisco packet tracer. Select 4 PC's onto workspace. Select & drag & one Switch.

Step 2: Cable the Devices

click connection → copper straight wire and connect all 4 PCs to Switch via Fast Ethernet ports.

Step 3: Assign IP address to PC's

Step 4: click the Switch → CLI tab. Enter following commands.

⇒ enable
⇒ configure terminal
⇒ interface vlan1
⇒ ip address 10.1.1.5 255.0.0.0
⇒ no shutdown
⇒ exit.

Step 5:- Verify connectivity using 'Ping'

click Pc → desktop → command prompt.

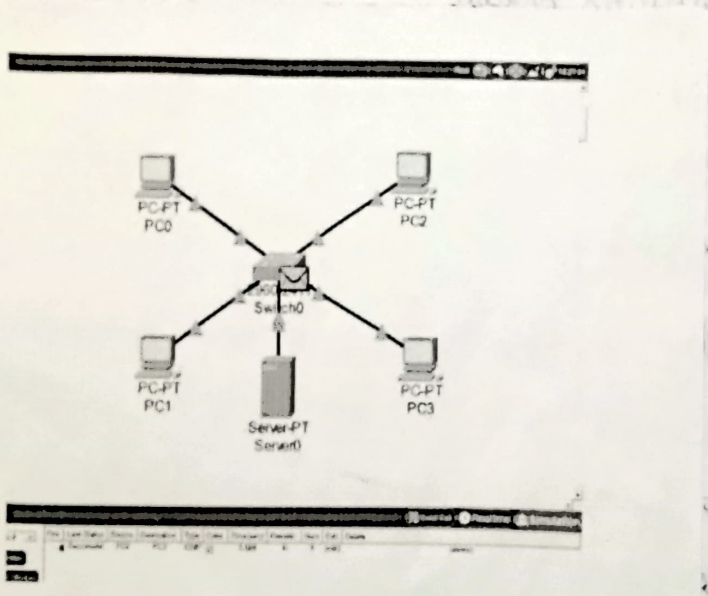
Use ping command to test connectivity

eg: Ping 10.1.1.2.

Step 6: Demonstrate Data Sharing

Add a service device and connect it to switch. Assign it up in: open the services server services.

Step 7: observe to simulation mode. Send a ping request btwn two device. Capture the step through packet as its travel hop by hop through switch. Observing mac address based forwarding.



Result:

Thus the setting up a local area Network (LAN) was implemented and executed successfully.